Functions & Graphs

ANSWER KEY

Section 1: Rational Functions & Oblique Asymptotes

1 Find the oblique asymptote of $f(x) = \frac{3x^2 - 2x + 4}{x - 1}$.

a. $y = 3x + 1$ ✓

b. $y = 3x - 1$

c. $y = 3x + 4$

d. $y = 3x$

Divide: $\frac{3x^2 - 2x + 4}{x - 1} = 3x + 1 + \frac{5}{x - 1}$. The oblique asymptote is $y = 3x + 1$.

2 Where does $g(x) = \frac{x^2 - 16}{x^2 - 3x - 4}$ have a vertical asymptote?

a. $x = -1$ ✓

b. $x = 4$

c. $x = -4$

d. Both $x = -1$ and $x = 4$

Factor: $\frac{(x - 4)(x + 4)}{(x - 4)(x + 1)}$. Common factor $(x - 4)$ cancels (hole at $x = 4$). The remaining denominator factor gives vertical asymptote at $x = -1$.

3 What is the horizontal asymptote of $h(x) = \frac{5x - 2}{x^2 + 1}$?

a. $y = 0$ ✓

b. $y = 5$

c. $y = -2$

d. No horizontal asymptote

Since the degree of the denominator (2) is strictly greater than the degree of the numerator (1), the horizontal asymptote is $y = 0$.

Section 2: Piecewise & Absolute Value Graphs

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4 What is the range of $f(x) = |2x - 6| - 4$?

a. $[-4, \infty)$ ✓

b. $[4, \infty)$

c. $[0, \infty)$

d. $(-\infty, -4]$

Since $|2x - 6| \geq 0$, the minimum value is $0 - 4 = -4$. Range is $[-4, \infty)$.

5 At what value of x does the graph of $y = |x - 3| + |x + 5|$ attain its minimum value?

a. **Any** $x \in [-5, 3]$ ✓

b. $x = -1$

c. $x = 3$

d. $x = -5$

For $x \in [-5, 3]$, $|x - 3| + |x + 5| = (3 - x) + (x + 5) = 8$. The constant minimum value 8 is achieved everywhere on $[-5, 3]$.

6 What is the domain of $f(x) = \sqrt{9 - x^2}$?

a. $[-3, 3]$ ✓

b. $(-\infty, -3] \cup [3, \infty)$

c. $[0, 3]$

d. $[-9, 9]$

Requirement: $9 - x^2 \geq 0 \Rightarrow x^2 \leq 9 \Rightarrow -3 \leq x \leq 3$.

Section 3: Inverse Trigonometric Graphs & Domains

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7 What is the domain of $f(x) = \arcsin(2x - 1)$?

a. $[0, 1]$ ✓

b. $[-1, 1]$

c. $[-0.5, 0.5]$

d. $[0, 2]$

Argument must be in $[-1, 1]$: $-1 \le 2x - 1 \le 1 \Rightarrow 0 \le 2x \le 2 \Rightarrow 0 \le x \le 1$.

8 What is the range of the principal $\arccos(x)$ function?

a. $[0, \pi]$ ✓

b. $[-\frac{\pi}{2}, \frac{\pi}{2}]$

c. $[0, 2\pi]$

d. $(0, \pi)$

By standard mathematical convention, the principal branch of $\arccos(x)$ has range $[0, \pi]$.

9 Evaluate $\cos(\arcsin(-\frac{4}{5}))$.

a. $\frac{3}{5}$ ✓

b. $-\frac{3}{5}$

c. $\frac{4}{5}$

d. $-\frac{4}{5}$

$\arcsin(-4/5)$ is in Quadrant IV where cosine is positive: $\cos\theta = \sqrt{1 - (-4/5)^2} = \frac{3}{5}$.

Section 4: Transformations of Non-Linear Functions

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10 How is the graph of $y = 2f(-x) + 3$ obtained from $y = f(x)$?

a. **Reflect over y-axis, vertical stretch by 2, shift up 3** ✓

b. Reflect over x-axis, vertical stretch by 2, shift up 3

c. Reflect over y-axis, horizontal stretch by 2, shift up 3

d. Reflect over origin, shift up 3

$f(-x)$ reflects across the y-axis; multiplying by 2 vertically stretches by 2; adding 3 shifts up by 3.

11 If the point $(2, 6)$ lies on $y = f(x)$, which point must lie on $y = f(2x) - 1$?

a. $(1, 5)$ ✓

b. $(4, 5)$

c. $(1, 7)$

d. $(4, 7)$

Horizontal compression by 2: x-coordinate becomes $\frac{2}{2} = 1$. Vertical shift down 1: y-coordinate becomes $6 - 1 = 5$. The point is $(1, 5)$.

12 Which transformation maps $y = x^3$ to $y = -(x - 1)^3 + 2$?

a. **Shift right 1, reflect over x-axis, shift up 2** ✓

b. Shift left 1, reflect over x-axis, shift up 2

c. Shift right 1, reflect over y-axis, shift down 2

d. Shift left 1, shift up 2

$(x - 1)$ shifts right 1; negation reflects across the x-axis; $+2$ shifts up 2.

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